

Simon Hathout Willard

hathoutwillard@gmail.com | (431)-293-4827 | hathoutwillard.github.io

Second year Engineering Physics student at UBC with skills in CAD, rapid prototyping, and engineering design. Strong computer skills and able to pick up new software and tools quickly.

TECHNICAL EXPERIENCE

Automation Subteam Member with Agroponics Design Team | 2023 - Present | UBC

- Used ESP32 and Arduino sensor systems to measure PH and temperature in a nutrient solution, improving plant growth and productivity
- Wrote code to format data using JSON and transfer it wirelessly using MQTT
- Communicated with other subteams to determine required data, and assisted in selection and procurement of appropriate sensors
- Used Git and Github for sharing code and documentation, improving maintainability

NorthForge Design Space | July-August 2024 | Winnipeg, MB

- Received formal training with woodshop and laser cutter tools, ensuring safe use
- Designed and fabricated personal projects using these tools including picture frames, miniature blinds, and a cross-stitch holder.
- Learned the basics of web development and used these skills to create a portfolio for my personal projects, which can be found at <https://hathoutwillard.github.io/posts/>
- Worked part time as a contractor with Maker Mind Toys, sanding and ensuring consistent quality of wooden toys for children.

SKILLS

Programming - Java, C, Python, HTML/CSS, Hugo SSG, Arduino

Design Tools - Fusion 360, Onshape, Machine Shop, 3D Printing, Laser Cutting, Woodshop

Software - Git and Github. Microsoft and Google suites, particularly Excel.

EDUCATION

2nd Year Engineering Physics | 2023 – Present | UBC Applied Sciences

- Courses include software development, engineering design, thermodynamics, and electric circuit design. A full list can be found at <https://hathoutwillard.github.io/courses/>
- Worked in a team of 6 to design and code a robotic claw capable of lifting a variety of objects, leading to the highest score in our section
- Learned how to safely and effectively use metal machining tools to create practical parts
- Cumulative average: 92.9%

OTHER EXPERIENCE

Cook at Camp Wasaga | July-August 2023 | Wasagaming, MB

- Lead a team to prepare meals for a camp of 60 without direct supervision, ensuring timely and high-quality service.
- Managed multiple concurrent tasks in a high stress environment. Implemented and refined existing recipes to better suit camp conditions.